Table 1: Comprehensive results on simulated scenarios at maximum depth $d = 2$.

Scenario	Train Accuracy (%)					Test Accuracy (%)				
	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS
Class_num = 2										
XOR	56.63 ± 4.68	56.63 ± 4.68	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	54.87 ± 4.89	54.87 ± 4.89	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26
DNF	78.48 ± 0.23	78.49 ± 0.23	79.03 ± 0.18	**79.04** ± 0.18	**79.04** ± 0.18	78.00 ± 0.55	78.06 ± 0.53	78.09 ± 0.48	**78.13** ± 0.48	**78.13** ± 0.48
Random Tree (d=3)	80.18 ± 2.86	80.18 ± 2.86	**83.63** ± 2.32	**83.63** ± 2.32	82.22 ± 2.17	**79.42** ± 2.98	**79.42** ± 2.98	82.94 ± 2.49	82.94 ± 2.49	81.33 ± 2.35
Random Tree (d=4)	72.79 ± 1.66	72.79 ± 1.66	**75.41** ± 1.49	**75.41** ± 1.49	**75.41** ± 1.49	72.00 ± 1.88	72.00 ± 1.88	74.76 ± 1.58	**74.77** ± 1.58	**74.77** ± 1.58
Linear	**83.79** ± 0.22	**83.79** ± 0.22	**83.79** ± 0.22	**83.79** ± 0.22	**83.79** ± 0.22	**83.76** ± 0.40	**83.76** ± 0.40	**83.76** ± 0.40	**83.76** ± 0.40	**83.76** ± 0.40
Circular	55.01 ± 0.14	55.01 ± 0.14	**55.91** ± 0.12	**55.91** ± 0.12	**55.91** ± 0.12	53.38 ± 0.62	53.38 ± 0.62	**53.43** ± 0.55	**53.43** ± 0.55	53.41 ± 0.56
Voting	68.08 ± 0.18	68.08 ± 0.18	68.80 ± 0.14	**68.81** ± 0.15	68.80 ± 0.14	**67.04** ± 0.43	**67.04** ± 0.43	66.93 ± 0.40	66.93 ± 0.40	67.01 ± 0.37
Class_num = 3										
XOR	51.08 ± 2.65	50.25 ± 2.20	**72.42** ± 0.17	**72.42** ± 0.17	**72.42** ± 0.17	51.18 ± 2.60	50.41 ± 2.21	**71.83** ± 0.47	**71.83** ± 0.47	**71.83** ± 0.47
DNF	78.15 ± 0.15	78.14 ± 0.16	**78.32** ± 0.14	**78.32** ± 0.14	**78.32** ± 0.14	77.71 ± 0.44	**77.78** ± 0.44	77.60 ± 0.46	77.60 ± 0.46	77.60 ± 0.46
Random Tree (d=3)	73.96 ± 2.92	74.38 ± 2.51	**76.82** ± 2.20	**76.83** ± 2.20	**76.83** ± 2.20	73.23 ± 2.94	73.81 ± 2.62	**76.56** ± 2.11	76.53 ± 2.10	76.53 ± 2.10
Random Tree (d=4)	62.90 ± 2.47	63.43 ± 2.30	**65.20** ± 1.88	**65.20** ± 1.88	**65.20** ± 1.88	62.67 ± 2.73	63.31 ± 2.38	**64.79** ± 2.03	**64.83** ± 2.04	**64.83** ± 2.04
Linear	**77.61** ± 0.23	**77.61** ± 0.23	**77.61** ± 0.23	**77.61** ± 0.23	**77.61** ± 0.23	**77.67** ± 0.53	**77.67** ± 0.53	**77.67** ± 0.53	**77.67** ± 0.53	**77.67** ± 0.53
Circular	40.93 ± 0.32	41.01 ± 0.30	**41.66** ± 0.19	**41.66** ± 0.20	**41.66** ± 0.20	40.18 ± 0.54	**40.26** ± 0.59	39.95 ± 0.65	40.00 ± 0.64	40.04 ± 0.64
Voting	61.84 ± 0.22	61.87 ± 0.20	61.99 ± 0.20	**62.00** ± 0.19	**62.00** ± 0.19	60.14 ± 0.46	60.13 ± 0.43	60.13 ± 0.41	**60.17** ± 0.41	60.16 ± 0.41

* *Note: Experimental settings are fixed at 20 features, sample size $N = 5000$, and noise level 0.05. Results are reported as Mean ± 95% Confidence Interval (CI) over 30 independent trials. The search budget for PG-DTS is $M = 2000$.*

Table 2: Comprehensive results on simulated scenarios at maximum depth $d = 3$.

Scenario	Train Accuracy (%)					Test Accuracy (%)				
	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS
Class_num = 2										
XOR	59.03 ± 4.75	59.03 ± 4.75	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	56.09 ± 5.15	56.09 ± 5.15	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26
DNF	89.72 ± 0.13	89.72 ± 0.13	**89.89** ± 0.14	**89.89** ± 0.14	**89.89** ± 0.14	**89.28** ± 0.29	89.24 ± 0.28	89.22 ± 0.34	89.21 ± 0.34	89.22 ± 0.34
Random Tree (d=3)	87.93 ± 2.73	87.93 ± 2.73	**95.07** ± 0.14	**95.07** ± 0.14	94.32 ± 1.02	87.55 ± 2.85	87.54 ± 2.85	**94.97** ± 0.27	**94.97** ± 0.27	94.24 ± 1.07
Random Tree (d=4)	79.94 ± 2.22	79.94 ± 2.22	**83.90** ± 1.54	**83.90** ± 1.54	83.59 ± 1.59	79.20 ± 2.34	79.20 ± 2.34	**83.38** ± 1.58	83.35 ± 1.58	82.91 ± 1.76
Linear	84.46 ± 0.18	84.48 ± 0.18	85.05 ± 0.15	**85.08** ± 0.14	84.75 ± 0.14	**83.92** ± 0.41	83.88 ± 0.39	83.72 ± 0.35	83.65 ± 0.33	83.79 ± 0.38
Circular	56.86 ± 0.20	56.86 ± 0.20	**58.13** ± 0.13	**58.13** ± 0.13	57.76 ± 0.17	54.14 ± 0.61	54.14 ± 0.61	54.00 ± 0.47	53.98 ± 0.47	**54.19** ± 0.59
Voting	79.18 ± 0.15	79.18 ± 0.15	**79.30** ± 0.15	**79.30** ± 0.15	**79.30** ± 0.15	**78.35** ± 0.36	**78.35** ± 0.36	78.30 ± 0.45	78.30 ± 0.45	78.30 ± 0.45
Class_num = 3										
XOR	51.89 ± 2.82	51.08 ± 2.46	**73.50** ± 0.17	**73.50** ± 0.17	**73.50** ± 0.17	51.95 ± 2.71	51.19 ± 2.38	**71.83** ± 0.39	**71.83** ± 0.39	**71.83** ± 0.39
DNF	89.52 ± 0.12	89.55 ± 0.12	**89.65** ± 0.13	**89.65** ± 0.12	**89.65** ± 0.12	**89.15** ± 0.29	89.07 ± 0.30	89.04 ± 0.31	89.05 ± 0.33	89.08 ± 0.32
Random Tree (d=3)	87.90 ± 2.65	85.07 ± 2.84	**95.01** ± 0.12	**95.01** ± 0.12	93.91 ± 1.15	87.44 ± 2.79	84.98 ± 2.87	**94.98** ± 0.22	**94.98** ± 0.22	93.74 ± 1.38
Random Tree (d=4)	72.70 ± 2.65	71.68 ± 2.53	**79.04** ± 2.07	**79.04** ± 2.07	78.48 ± 2.09	72.20 ± 2.56	71.06 ± 2.57	**78.72** ± 2.08	78.65 ± 2.09	78.13 ± 2.11
Linear	78.64 ± 0.15	78.70 ± 0.15	79.16 ± 0.19	**79.18** ± 0.19	79.01 ± 0.20	77.34 ± 0.49	77.41 ± 0.47	77.68 ± 0.54	77.74 ± 0.59	**77.98** ± 0.58
Circular	42.22 ± 0.25	42.32 ± 0.25	43.68 ± 0.18	**43.69** ± 0.18	43.43 ± 0.20	40.19 ± 0.54	40.21 ± 0.65	40.92 ± 0.53	40.85 ± 0.62	**41.02** ± 0.54
Voting	73.18 ± 0.21	73.16 ± 0.20	**73.22** ± 0.21	**73.22** ± 0.21	73.21 ± 0.21	71.98 ± 0.61	71.95 ± 0.55	**72.19** ± 0.47	**72.19** ± 0.47	72.18 ± 0.48

* *Note: Experimental settings are fixed at 20 features, sample size $N = 5000$, and noise level 0.05. Results are reported as Mean ± 95% Confidence Interval (CI) over 30 independent trials. The search budget for PG-DTS is $M = 4000$.*

Table 3: Comprehensive results on simulated scenarios at maximum depth $d = 4$.

Scenario	Train Accuracy (%)					Test Accuracy (%)				
	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS
Class_num = 2										
XOR	64.21 ± 4.48	64.21 ± 4.48	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	60.97 ± 5.02	60.97 ± 5.02	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26
DNF	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26
Random Tree (d=3)	91.74 ± 2.13	91.74 ± 2.13	**95.07** ± 0.14	**95.07** ± 0.14	**95.07** ± 0.14	91.27 ± 2.30	91.27 ± 2.30	**94.97** ± 0.27	**94.97** ± 0.27	**94.97** ± 0.27
Random Tree (d=4)	85.91 ± 1.63	85.91 ± 1.63	**95.03** ± 0.13	**95.03** ± 0.13	92.87 ± 1.25	85.14 ± 1.69	85.18 ± 1.70	**95.00** ± 0.29	**95.00** ± 0.29	92.76 ± 1.30
Linear	**95.03** ± 0.14	**95.03** ± 0.14	**95.03** ± 0.14	**95.03** ± 0.14	**95.03** ± 0.14	**95.04** ± 0.26	**95.04** ± 0.26	**95.04** ± 0.26	**95.04** ± 0.26	**95.04** ± 0.26
Circular	59.54 ± 0.21	59.54 ± 0.21	61.35 ± 0.12	**61.38** ± 0.11	60.06 ± 0.18	**56.03** ± 0.64	**56.03** ± 0.64	55.62 ± 0.63	55.53 ± 0.63	55.67 ± 0.58
Voting	79.52 ± 0.17	79.53 ± 0.18	80.59 ± 0.12	**80.67** ± 0.12	80.09 ± 0.14	**78.33** ± 0.45	**78.33** ± 0.46	78.29 ± 0.49	78.25 ± 0.56	78.23 ± 0.47
Class_num = 3										
XOR	54.08 ± 2.88	53.73 ± 2.70	74.79 ± 0.18	**74.80** ± 0.17	74.75 ± 0.18	54.01 ± 2.77	53.51 ± 2.55	71.79 ± 0.30	**71.87** ± 0.30	71.77 ± 0.38
DNF	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	**95.02** ± 0.11	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26	**94.86** ± 0.26
Random Tree (d=3)	93.40 ± 1.17	91.73 ± 1.86	**95.01** ± 0.12	**95.01** ± 0.12	**95.01** ± 0.12	93.34 ± 1.19	91.54 ± 1.95	**94.98** ± 0.22	**94.98** ± 0.22	**94.98** ± 0.22
Random Tree (d=4)	83.25 ± 2.51	82.54 ± 2.25	**95.03** ± 0.13	**95.03** ± 0.13	93.75 ± 0.99	82.64 ± 2.63	82.12 ± 2.26	**94.86** ± 0.24	**94.86** ± 0.24	93.43 ± 1.05
Linear	**95.03** ± 0.14	**95.03** ± 0.14	**95.03** ± 0.14	**95.03** ± 0.14	**95.03** ± 0.14	**95.04** ± 0.26	**95.04** ± 0.26	**95.04** ± 0.26	**95.04** ± 0.26	**95.04** ± 0.26
Circular	44.90 ± 0.19	44.93 ± 0.19	46.31 ± 0.13	**46.36** ± 0.14	45.08 ± 0.23	41.75 ± 0.62	41.61 ± 0.63	41.58 ± 0.61	41.06 ± 0.55	**41.86** ± 0.59
Voting	73.77 ± 0.20	73.84 ± 0.19	74.98 ± 0.17	**75.08** ± 0.16	74.17 ± 0.19	72.06 ± 0.52	**72.16** ± 0.52	72.10 ± 0.55	72.11 ± 0.55	71.92 ± 0.41

* *Note: Experimental settings are fixed at 20 features, sample size $N = 5000$, and noise level 0.05. Results are reported as Mean ± 95% Confidence Interval (CI) over 30 independent trials. The search budget for PG-DTS is $M = 8000$.*

Table 4: Average runtime (sec) on simulated scenarios across different depths and budgets.

Setting	Average Runtime (seconds)		
	$d = 2, M = 2000$	$d = 3, M = 4000$	$d = 4, M = 8000$
Class_num = 2	0.2971	3.9811	19.1000
Class_num = 3	0.3174	3.5872	18.2923

* *Note: Runtimes are averaged over all scenarios within the same class number and depth setting. M denotes the number of search iterations for PG-DTS.*

Table 5: Comprehensive results on datasets at maximum depth $d = 2$.

Dataset	Train Accuracy (%)					Test Accuracy (%)					Runtime (sec)					Number of Leaves				
	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS
balance-scale	69.04	70.12	68.92	69.00	**72.52**	66.08	**69.12**	65.12	64.80	68.32	0.001	0.001	0.004	0.001	0.090	4.0	4.0	3.4	3.8	4.0
breast-cancer	93.38	94.67	94.78	94.78	**96.42**	91.84	93.85	93.85	93.85	**96.14**	0.001	0.001	0.018	0.004	0.605	4.0	4.0	4.0	4.0	4.0
credit-approval	85.51	85.72	**86.30**	**86.30**	86.16	**85.51**	85.22	83.62	83.62	84.93	0.002	0.002	0.012	0.002	0.163	4.0	4.0	4.0	4.0	4.0
credit-g	71.80	71.80	**73.43**	**73.43**	73.20	70.60	70.60	70.30	70.30	**71.40**	0.002	0.002	0.019	0.002	0.740	4.0	4.0	3.8	3.8	4.0
diabetes	76.59	**76.63**	75.91	75.91	76.43	73.44	73.57	73.44	73.44	**76.30**	0.001	0.001	0.006	0.001	0.157	4.0	4.0	4.0	4.0	4.0
hayes-roth	56.88	56.88	65.31	**65.47**	64.38	46.25	46.25	50.00	50.62	**54.38**	0.001	0.001	0.002	0.001	0.023	3.0	3.0	3.8	4.0	4.0
lymphography	78.04	78.38	82.77	**82.94**	82.26	72.90	70.85	73.61	73.61	**74.25**	0.001	0.001	0.008	0.001	0.130	4.0	4.0	3.8	4.0	4.0
monk-1	73.39	73.39	**82.26**	**82.26**	**82.26**	**75.00**	**75.00**	72.22	72.22	72.22	0.001	0.001	0.005	0.003	0.188	3.0	3.0	4.0	4.0	4.0
monk-2	63.91	63.91	**66.27**	**66.27**	**66.27**	**62.50**	**62.50**	60.65	60.65	60.65	0.001	0.001	0.005	0.002	0.173	4.0	4.0	4.0	4.0	4.0
monk-3	**93.44**	**93.44**	**93.44**	**93.44**	**93.44**	**97.22**	**97.22**	**97.22**	**97.22**	**97.22**	0.001	0.001	0.004	0.002	0.085	4.0	4.0	3.0	3.0	4.0
sonar	79.69	**81.37**	78.25	78.25	80.53	69.23	64.88	**74.55**	**74.55**	68.73	0.003	0.003	0.193	0.012	0.777	4.0	4.0	4.0	4.0	4.0
tic-tac-toe	70.67	70.67	71.11	**71.14**	**71.14**	67.43	67.43	66.60	**67.53**	**67.53**	0.001	0.001	0.006	0.001	0.163	4.0	4.0	3.0	3.2	4.0
voting	95.46	95.63	95.98	**96.32**	95.98	95.17	**95.63**	94.94	94.48	95.17	0.001	0.001	0.008	0.002	0.228	3.8	4.0	2.8	4.0	4.0
zoo	75.73	73.27	**82.92**	**82.92**	**82.92**	**72.24**	73.24	**75.14**	**75.14**	**75.14**	0.001	0.001	0.004	0.001	0.084	3.2	3.0	4.0	4.0	4.0
adult	82.72	82.72	-	80.52	**82.74**	82.67	82.67	-	80.52	**82.68**	0.086	0.088	0.461	0.094	12.401	4.0	4.0	4.0	4.0	4.0
car	**77.78**	**77.78**	**77.78**	**77.78**	**77.78**	**77.78**	**77.78**	**77.78**	**77.78**	**77.78**	0.001	0.001	0.006	0.001	0.096	3.0	3.0	3.0	3.0	3.0
chess	24.13	23.97	24.12	24.12	**25.73**	23.92	23.84	23.95	23.95	**25.73**	0.010	0.010	0.212	0.031	0.537	4.0	4.0	4.0	4.0	4.0
connect-4	65.83	67.26	65.83	**67.27**	**67.27**	65.83	**67.26**	65.83	67.25	67.25	0.101	0.119	0.812	0.158	52.581	4.0	4.0	1.0	4.0	4.0
mushroom	92.71	95.42	**96.90**	**96.90**	**96.90**	92.71	95.42	**96.90**	**96.90**	**96.90**	0.011	0.011	0.072	0.013	0.977	4.0	4.0	4.0	4.0	4.0
nursery	**82.52**	**82.52**	76.34	76.34	**82.52**	**82.52**	**82.52**	76.34	76.34	**82.52**	0.004	0.005	0.032	0.007	0.041	3.0	3.0	3.0	3.0	3.0

** Note: All datasets are evaluated using 5-fold cross-validation with an 8:2 train/test split. For PG-DTS, the search budget is set to $M = 500$ for the upper block of datasets (up to zoo), and $M = 1000$ for the lower block of larger datasets (from adult downwards). "OOM" indicates Out-Of-Memory.*

Table 6: Comprehensive results on datasets at maximum depth $d = 3$.

Dataset	Train Accuracy (%)					Test Accuracy (%)					Runtime (sec)					Number of Leaves				
	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS
balance-scale	76.08	76.68	75.00	75.04	**78.44**	70.72	72.16	68.64	68.80	**73.44**	0.001	0.001	0.042	0.004	0.727	8.0	8.0	7.6	7.8	8.0
breast-cancer	96.46	96.42	96.35	96.42	**97.42**	93.85	94.56	93.99	93.71	**95.28**	0.001	0.001	1.360	0.091	4.143	8.0	7.8	7.0	7.8	7.0
credit-approval	86.09	86.56	**87.93**	**88.08**	87.68	**84.78**	83.91	83.04	83.62	84.49	0.002	0.002	0.747	0.063	2.683	7.8	8.0	7.2	8.0	6.4
credit-g	73.80	74.10	**76.45**	**76.45**	76.17	70.80	70.70	72.70	72.70	**75.80**	0.002	0.002	2.057	0.140	4.194	8.0	8.0	8.0	8.0	8.0
diabetes	77.15	77.60	78.45	78.45	**78.52**	**74.35**	73.31	73.18	73.18	73.69	0.002	0.002	0.211	0.017	1.565	8.0	8.0	8.0	8.0	7.4
hayes-roth	61.09	61.09	**79.69**	**79.69**	71.72	54.38	54.38	**64.38**	**64.38**	57.50	0.001	0.001	0.008	0.001	0.216	4.0	4.0	8.0	8.0	6.8
lymphography	81.92	86.82	89.52	**90.20**	88.17	73.59	72.94	**76.32**	**76.32**	72.90	0.001	0.001	0.304	0.016	1.180	7.6	7.8	7.2	8.0	8.0
monk-1	91.13	91.13	91.13	**91.94**	91.13	83.33	83.33	83.33	**88.89**	83.33	0.001	0.001	0.018	0.003	0.117	5.0	5.0	5.0	7.0	5.0
monk-2	66.86	68.64	**75.74**	**75.74**	**75.74**	**66.20**	65.74	63.66	63.66	62.73	0.001	0.001	0.018	0.004	0.629	8.0	8.0	7.0	7.0	8.0
monk-3	94.26	**95.08**	93.44	**95.08**	**95.08**	94.44	**100.00**	97.22	**100.00**	**100.00**	0.001	0.001	0.016	0.003	0.320	6.0	6.0	3.0	6.0	7.0
sonar	86.17	**88.94**	86.90	87.02	87.98	69.77	**73.07**	71.66	71.66	67.34	0.004	0.003	100.121	3.250	3.968	7.4	7.8	7.8	8.0	8.0
tic-tac-toe	75.39	75.39	**78.58**	**78.58**	78.13	74.22	74.22	72.76	72.76	**74.64**	0.001	0.001	0.090	0.007	1.609	8.0	8.0	7.8	7.8	8.0
voting	96.78	96.90	96.78	**97.53**	97.13	94.48	94.02	94.48	94.48	**94.71**	0.001	0.001	0.341	0.021	2.459	6.6	7.6	4.0	7.0	7.6
zoo	87.36	83.91	94.06	**95.29**	95.05	78.14	78.14	83.10	**90.00**	86.10	0.001	0.001	0.044	0.002	0.538	4.8	4.0	6.2	8.0	8.0
adult	82.72	**83.14**	82.52	82.56	84.04	82.67	**83.06**	82.41	82.56	84.03	0.122	0.120	43.567	3.386	94.507	8.0	8.0	7.4	8.0	8.0
car	79.17	79.17	81.44	81.44	**81.93**	79.17	79.17	79.92	79.92	**80.85**	0.001	0.001	0.061	0.006	1.547	4.0	4.0	5.6	6.0	6.8
chess	28.23	27.84	27.15	27.15	**28.67**	28.15	27.82	26.90	26.90	**28.55**	0.013	0.013	8.917	0.730	6.458	8.0	8.0	8.0	8.0	8.0
connect-4	66.43	68.17	65.83	**68.90**	68.41	66.39	68.06	65.83	**68.90**	68.27	0.146	0.150	79.195	9.913	327.305	8.0	8.0	1.0	8.0	6.8
mushroom	96.16	98.54	**99.90**	**99.90**	**99.90**	96.16	98.44	**99.90**	**99.90**	**99.90**	0.013	0.014	4.404	0.300	11.647	6.0	7.0	7.0	7.0	7.0
nursery	84.77	84.77	83.77	83.77	**85.92**	84.77	84.77	83.77	83.77	**85.73**	0.006	0.006	0.387	0.043	0.427	5.0	5.0	7.0	7.0	5.0

** Note: All datasets are evaluated using 5-fold cross-validation with an 8:2 train/test split. For PG-DTS, the search budget is set to $M = 1000$ for the upper block of datasets (up to zoo), and $M = 3000$ for the lower block of larger datasets (from adult downwards).*

Table 7: Comprehensive results on datasets at maximum depth $d = 4$.

Dataset	Train Accuracy (%)					Test Accuracy (%)					Runtime (sec)					Number of Leaves				
	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS	DT-Ent	CART	GOSDT	DL8.5	PG-DTS
balance-scale	80.92	**83.52**	78.68	79.28	83.24	72.96	77.60	70.56	71.36	**79.36**	0.001	0.001	0.517	0.032	2.597	15.6	16.0	12.0	15.2	11.2
breast-cancer	97.39	97.43	97.28	97.93	**97.96**	**95.42**	94.85	94.42	93.85	94.71	0.001	0.001	101.748	3.140	11.867	12.8	13.4	9.4	15.0	9.4
credit-approval	88.55	89.64	89.46	**90.51**	89.28	82.75	83.33	83.33	82.75	**84.20**	0.002	0.002	34.659	2.077	10.404	13.4	14.4	9.6	16.0	9.0
credit-g	75.75	76.82	79.72	**79.78**	77.92	70.20	71.10	72.00	72.30	**74.80**	0.003	0.003	128.949	7.039	14.097	15.0	15.2	15.6	16.0	13.4
diabetes	78.97	80.63	81.02	**81.31**	79.95	74.48	73.96	72.66	73.30	**75.13**	0.002	0.002	6.692	0.393	5.767	14.6	15.2	13.8	16.0	10.4
hayes-roth	72.50	72.50	86.88	**87.81**	83.75	66.88	66.88	70.62	70.00	**74.38**	0.001	0.001	0.042	0.003	0.682	5.0	5.0	9.6	12.2	11.2
lymphography	88.18	91.04	94.42	**97.13**	92.40	74.32	74.94	80.30	**81.06**	74.94	0.001	0.001	7.506	0.243	4.219	11.6	12.2	10.8	16.0	13.2
monk-1	91.13	91.13	**100.00**	**100.00**	**100.00**	83.33	83.33	**100.00**	**100.00**	**100.00**	0.001	0.001	0.161	0.003	0.652	6.0	6.0	7.0	11.0	8.0
monk-2	69.82	72.19	80.47	**82.25**	79.29	64.81	64.35	65.74	**66.20**	65.28	0.001	0.001	0.191	0.011	2.575	15.0	16.0	11.0	15.0	12.0
monk-3	95.90	95.08	93.44	**98.36**	95.90	95.37	**100.00**	97.22	90.74	95.37	0.001	0.001	0.128	0.006	1.560	9.0	9.0	3.0	15.0	7.0
sonar	92.54	**95.91**	-	94.11	93.15	69.21	71.15	-	71.72	**78.35**	0.005	0.004	-	680.365	25.395	11.0	12.8	-	15.6	14.2
tic-tac-toe	84.32	84.32	86.61	**86.64**	83.59	**81.11**	**81.11**	80.27	80.17	77.14	0.002	0.002	2.010	0.101	4.954	14.0	14.0	13.2	15.0	13.4
voting	97.59	97.99	97.93	**99.25**	98.28	95.63	95.63	**96.32**	93.33	95.63	0.001	0.001	8.986	0.333	7.896	10.8	12.4	5.8	15.0	9.8
zoo	95.79	89.60	96.78	**100.00**	99.01	86.10	78.14	87.10	88.00	**88.05**	0.001	0.001	0.390	0.003	1.665	6.6	5.0	6.6	11.8	9.4
adult	83.53	83.86	-	83.91	**84.43**	83.48	83.74	-	83.68	**84.31**	0.157	0.154	-	166.813	236.532	15.6	15.8	-	15.8	10.4
car	84.09	84.85	84.95	85.01	**87.15**	83.10	84.20	83.22	83.22	**87.15**	0.001	0.001	0.887	0.059	3.161	6.0	6.0	10.4	12.2	8.6
chess	30.78	30.40	30.77	30.77	**32.41**	30.73	30.05	30.50	30.50	**32.38**	0.017	0.017	248.983	16.904	18.838	16.0	16.0	16.0	16.0	14.0
connect-4	67.42	69.13	-	**70.52**	69.48	67.36	69.02	-	**70.52**	69.38	0.184	0.182	-	626.498	495.106	16.0	16.0	-	16.0	10.8
mushroom	98.76	99.38	99.95	**100.00**	**100.00**	98.76	99.11	99.91	**100.00**	**100.00**	0.015	0.018	568.745	1.950	32.910	9.0	9.8	7.4	16.0	10.8
nursery	85.88	85.88	88.78	**88.79**	87.82	85.88	85.88	**88.65**	88.58	87.82	0.006	0.006	7.447	0.469	6.528	9.0	9.0	13.0	15.0	8.0

** Note: All datasets are evaluated using 5-fold cross-validation with an 8:2 train/test split. For PG-DTS, the search budget is set to $M = 2000$ for the upper block of datasets (up to zoo), and $M = 5000$ for the lower block of larger datasets (from adult downwards). "-" indicates that the model size is not applicable due to OOM.*